

HW 8:**Prob 1:**

$$a)\beta = 46.51$$

$$b)\beta = 8307$$

$$c)\beta = 71.53$$

Prob 2:

$$V_2 = 1.218V$$

$$V_1 = -0.913V$$

$$i_4 = 486\mu A$$

$$V_3 = 1.258V$$

$$V_5 = -0.894V$$

$$V_6 = 0.191V$$

$$V_7 = 1.4016V$$

Prob 4:

$$a)V_1 = 0V$$

$$V_2 = 0.7V$$

$$V_3 = -0.7V$$

$$V_4 = -1.4V$$

$$V_5 = 1.10V$$

$$b)V_1 = 0.226V$$

$$V_2 = 0.926V$$

$$V_3 = -0.974V$$

$$V_4 = -1.674V$$

$$V_5 = 1.445V$$

Prob 5:

+40dB/dec

20dB

-90 degree

Prob 6:

$$a) i_E = 2.795 \mu A$$

$$i_B = 9.285 \mu A$$

$$i_C = 2.785 \mu A$$

$$V_B = 2.0975 V$$

$$V_C = 9.215 V$$

$$V_E = 1.3975 V$$

$$b) i_E = 18.29 mA$$

$$i_B = 60.76 \mu A$$

$$i_C = 18.23 mA$$

$$V_B = 9.845 V$$

$$V_C = -6.23 V$$

$$V_E = 9.14$$

$$c) i_E = 664.3 \mu A$$

$$i_B = 2.2 \mu A$$

$$i_C = 662 \mu A$$

$$V_B = 9.84 V$$

$$V_C = -10.21 V$$

$$V_E = 10.53 V$$

$$d) i_E = 1.63 mA$$

$$i_B = 5.414 \mu A$$

$$i_C = 1.624 mA$$

$$V_B = 4.997 V$$

$$V_C = -4.243 V$$

$$V_E = 4.297 V$$

Prob 7:

$$a) i_E = 2.603mA$$

$$i_B = 17.24\mu A$$

$$i_C = 2.586mA$$

$$V_B = 8.62mV$$

$$V_C = 3.631V$$

$$V_E = 0.709V$$

$$b) i_E = -161.57mA$$

$$i_B = -1.07mA$$

$$i_C = -160.5mA$$

$$V_B = 0.7V$$

$$V_C = 170.5V$$

$$V_E = 0V$$

$$c) i_E = 140.43mA$$

$$i_B = 930\mu A$$

$$i_C = 139.5mA$$

$$V_B = 0.7V$$

$$V_C = -129.5V$$

$$V_E = 0V$$